

Question 1

sum, differences ,Product and Quotient

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int a, b;
```

```
    cout << "Enter two integers: ";
```

```
    cin >> a >> b;
```

```
    cout << "Sum = " << a + b << endl;
```

```
    cout << "Difference = " << a - b << endl;
```

```
    cout << "Product = " << a * b << endl;
```

```
    cout << "Quotient = " << (float)a / b << endl;
```

```
    return 0;
```

```
}
```

Question 2

Area of a circle

```
#include <iostream>
```

```
#include <cmath>
```

```
using namespace std;
```

```
int main() {
```

```
    double radius, area;
```

```
    const double PI = 3.14159;
```

```
cout << "Enter radius: ";  
cin >> radius;  
  
area = PI \* pow(radius, 2);  
  
cout << "Area = " << area << endl;  
  
return 0;  
}
```

Question 3
Checking if integers are equals
#include<iostream>

```
using namespace std;  
  
int main() {  
    int a, b, c;  
  
    cout << "Enter three integers: ";  
    cin >> a >> b >> c;  
  
    if (a == b && b == c)  
        cout << "Equal";  
    else  
        cout << "Not equal";  
  
    return 0;  
}
```

Question 4

Check if string is 'Hello'

```
#include <iostream>

#include <string>

using namespace std;

int main() {
    string text;

    cout << "Enter a string: ";
    cin >> text;

    if (text == "hello")
        cout << "Hello!";
    else
        cout << "Goodbye!";

    return 0;
}
```

Question 5

Comparing 2 integers

```
#include <iostream>

using namespace std;

int main() {
    int a, b;

    cout << "Enter two integers: ";
    cin >> a >> b;
```

```
    if (a > b)
        cout << "First value is greater";
    else
        cout << "Second value is greater";

    return 0;
}
```

Question 6

check uppercase letter

```
#include <iostream>
#include <cctype>
using namespace std;

int main() {
    char ch;

    cout << "Enter a character: ";
    cin >> ch;

    if (isupper(ch))
        cout << "Uppercase letter";
    else
        cout << "Not an uppercase letter";

    return 0;
}
```

Question 7

check if strings contains 'a'

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    string text;
```

```
    cout << "Enter a string: ";
```

```
    cin >> text;
```

```
    if (text.find('a') != string::npos)
```

```
        cout << "Contains 'a'";
```

```
    else
```

```
        cout << "Does not contain 'a'";
```

```
    return 0;
```

```
}
```

Question 8

check odd or even

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int num;
```

```
    cout << "Enter a number: ";
```

```
cin >> num;

if (num % 2 != 0)
    cout << "Odd number";
else
    cout << "Even number";

return 0;
}
```

Question 9
using the ternary operators to determine the larger

```
#include <iostream>
using namespace std;

int main() {
    int a, b;

    cout << "Enter two integers: ";
    cin >> a >> b;

    cout << "Larger number = " << (a > b ? a : b);

    return 0;
}
```

Question 10
Checking if a number is both odd and divisible by 3 using the logical AND operator

```
#include <iostream>
using namespace std;
```

```
int main() {  
    int num;  
    cout << "Enter a number: ";  
    cin >> num;  
  
    if (num % 2 != 0 && num % 3 == 0)  
        cout << "Number is odd and divisible by 3";  
    else  
        cout << "Condition not satisfied";  
  
    return 0;  
}
```

Question 11

using the bitwise XOR operator to swap the values of 2 variables

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {  
    int a, b;  
  
    cout << "Enter two numbers: ";  
    cin >> a >> b;  
  
    a = a ^ b;  
    b = a ^ b;  
    a = a ^ b;
```

```
cout << "After swapping:" << endl;
cout << "a = " << a << endl;
cout << "b = " << b << endl;

return 0;
}
```

Question 12

determining the larger of two floating point numbers that uses the conditional operation

```
#include <iostream>
using namespace std;
```

```
int main() {
    float a, b;

    cout << "Enter two floating-point numbers: ";
    cin >> a >> b;

    cout << "Larger number = " << (a > b ? a : b);

    return 0;
}
```

Question 13

check if a number is a power of 2

```
#include <iostream>
using namespace std;
```

```
int main() {
    int num;
```



```
cout << "Enter a number: ";  
cin >> num;  
  
if (num > 0 && (num & (num - 1)) == 0)  
    cout << "Power of 2";  
else  
    cout << "Not a power of 2";  
  
return 0;  
}
```

Question 14

Binary to decimal conversion

```
#include <iostream>  
  
using namespace std;  
  
int main() {  
    long binary;  
    int decimal = 0, base = 1, rem;  
  
    cout << "Enter a binary number: ";  
    cin >> binary;  
  
    while (binary > 0) {  
        rem = binary % 10;  
        decimal += rem * base;  
        binary /= 10;  
        base *= 2;  
    }
```

```
}

cout << "Decimal = " << decimal;

return 0;
}
```

Question 15

Binary to decimal, Binary to hexadecimal and Binary to octadecimal

```
#include <iostream>

#include <bitset>

using namespace std;

int main() {
    string binary;
    int decimal = 0;

    cout << "Enter a binary number: ";
    cin >> binary;

    for (char bit : binary) {
        decimal = decimal * 2 + (bit - '0');
    }

    cout << "Decimal = " << decimal << endl;
    cout << "Octal = " << oct << decimal << endl;
    cout << "Hexadecimal = " << hex << decimal << endl;

    return 0;
}
```

}

